

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time work authorization, bringing over 4 years of experience in software development and data engineering, specializing in building scalable microservices and distributed data pipelines. Possesses a Master of Science in Computer Science with expertise in Java, Python, Spark, SQL, AWS, and Azure, coupled with experience in CI/CD automation and large-scale data platforms. Eager to leverage analytical skills and system architecture knowledge as an AI Platform Backend Engineer.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Engineered high-performance REST and SOAP API microservices using Java and Spring Boot, managing over 50,000 daily requests with 99.99% uptime and adhering to strict SLAs.
- Automated CI/CD workflows utilizing Jenkins and Azure DevOps, facilitating continuous delivery across cloud environments and reducing deployment cycles by 25%.
- Modernized legacy systems by migrating to Log4j 2.x, enhancing monitoring capabilities and decreasing issue resolution time by 20% through improved error handling.
- Spearheaded an EDX platform upgrade, streamlining software design and dependency management to achieve 50% faster delivery timelines.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions, employing Git for version control and JUnit for robust testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Developed enterprise applications and Restful Api services in Java and EJB for automotive clients, optimizing performance by 90% through SQL query tuning and code modernization.
- Implemented CI/CD pipelines with Jenkins and GitLab, incorporating automated testing and quality gates to ensure software design integrity.
- Mentored junior developers by conducting code reviews and debugging sessions, promoting clean coding practices and enhancing overall team productivity.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99% uptime through proactive monitoring and performance optimization workflows.
- Enhanced SQL data pipelines and database operations, improving query performance by 10% and reinforcing data integrity for critical business functions.
- Implemented comprehensive unit testing using JUnit4, achieving 99% code coverage and ensuring robust software development.
- Supported release planning and sprint execution using JIRA within Agile development teams, contributing to efficient product management.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Terraform) January 2024 - May 2025

- Built a distributed Spark-based data pipeline for near real-time processing and analysis of over 8 GB of NYC taxi trip data, demonstrating strong ETL pipeline capabilities.
- Developed an interactive Flask web application for data exploration, featuring dynamic filtering and visualization tools to support analytical skills.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments and scalable system architecture.
- Optimized Spark memory usage and partitioning, reducing overall job runtime by 40% and showcasing performance optimization.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight for infrastructure automation.

Privacy-Preserving AI Email Client (Capstone Project) (Python, Hugging Face, T5, LoRA) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, showcasing expertise in AI platform backend development.
- Applied parameter-efficient fine-tuning using LoRA to support low-resource training under privacy constraints, integrating advanced machine learning techniques.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment, emphasizing privacy-preserving AI applications.

SKILLS

- Programming Languages: Proficient in Java, Python, and SQL for robust software development and data manipulation.
- Frameworks & Libraries: Extensive experience with Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, and LoRA for building scalable AI platforms and data workflows.

- Web Services & APIs: Skilled in developing and consuming RESTful Api and SOAP API, ensuring seamless integration and efficient data exchange.
- Cloud Platforms: Expertise in AWS (EC2, RDS, S3, Lambda) and Azure (Functions, DevOps, ServiceBus) for cloud-native application development and infrastructure automation.
- Databases: Experienced with PostgreSQL, Oracle Database, IBM DB2, and RDS, focusing on data structure and efficient data management.
- Tools & Technologies: Proficient in Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, Tableau, DBeaver, and Terraform for efficient software development lifecycle and system architecture.
- Core Concepts: Strong understanding of ETL Pipelines, Data Warehousing, Data Analytics, Agile, Scrum, Test Driven Development, Object Oriented Programming, CI/CD, Microservices, Large-scale data platforms, Infrastructure automation, and Privacy-preserving LLM-based applications.
- Analytical & Soft Skills: Demonstrated analytical skills, product management acumen, system architecture design, and effective problem-solving abilities.